Lab Report: Model comparison on benchmarks

1 Introduction

In this report we compare several models on standard benchmarks. The dataset was split into 80 percent training data and the rest for testing. Each experiment was repeated 3 times with different random seeds.

2 Results

Figure 1 shows the accuracy of ResNet-50 by dataset.

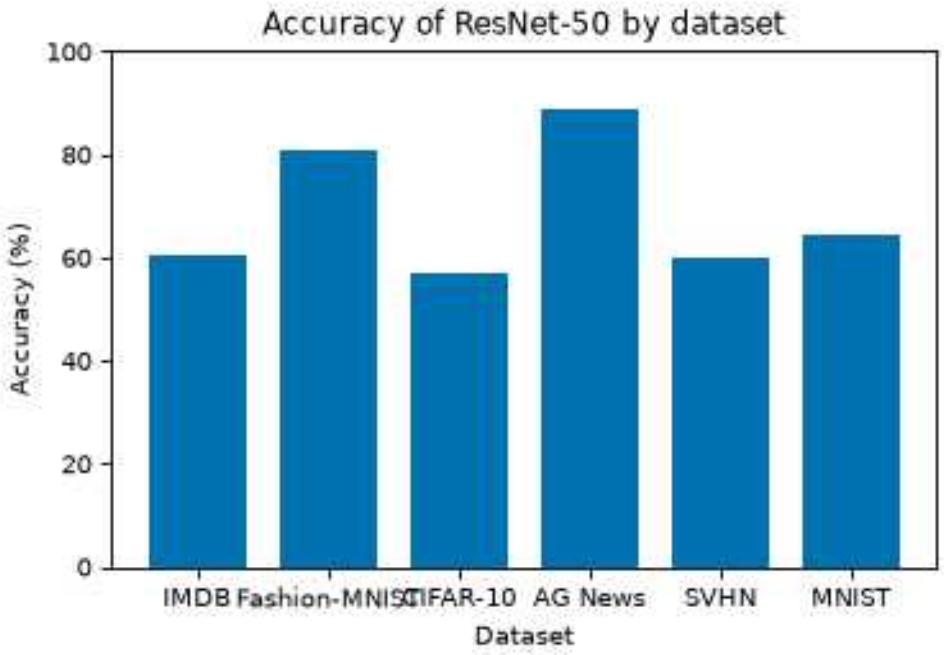


Figure 1: Accuracy of ResNet-50 by dataset for the models in this experiment.

ResNet-50 scores 24.5 points higher on AG News than on MNIST. ResNet-50 achieves over 75% accuracy on Fashion-MNIST. On IMDB, ResNet-50 obtains a accuracy of over 55% (Figure 1).

Figure 2 shows the training loss.

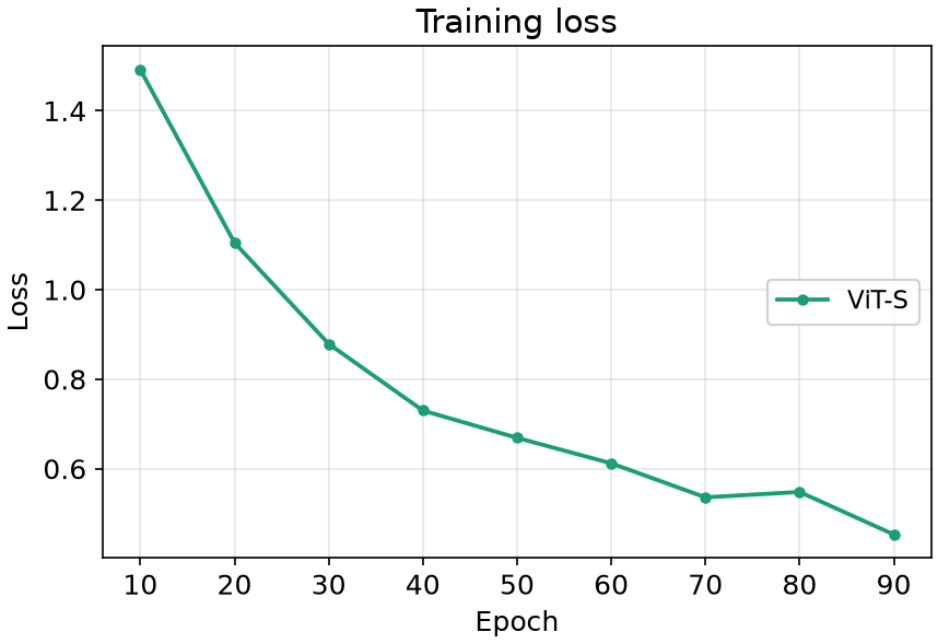


Figure 2: Training loss for the models in this experiment.

At epoch 30, the loss of ViT-S is 0.88. At epoch 60, the loss of ViT-S is 0.669. At epoch 60, the loss of ViT-S is about 0.6.

Figure 3 shows the test accuracy during training.

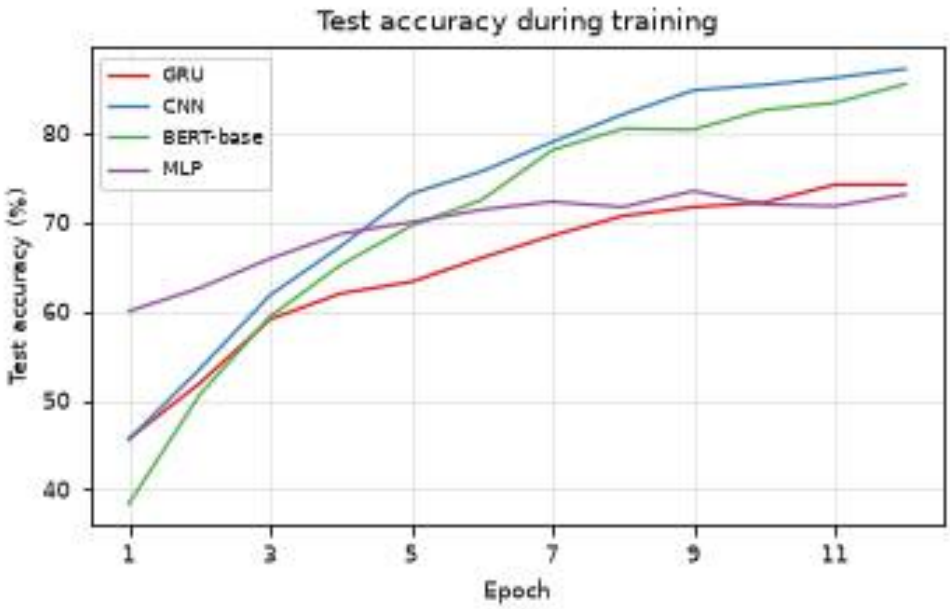


Figure 3: Test accuracy during training for the models in this experiment.

From epoch 4 to epoch 11, the test accuracy of CNN improves by a relative 20.1%. CNN attains a test accuracy of 81.8% at epoch 12. CNN reaches 73.2% test accuracy at epoch 5.

3 Discussion

On STL-10, Naive Bayes obtains an accuracy of 64.0%. We note that the training was stable for all models.